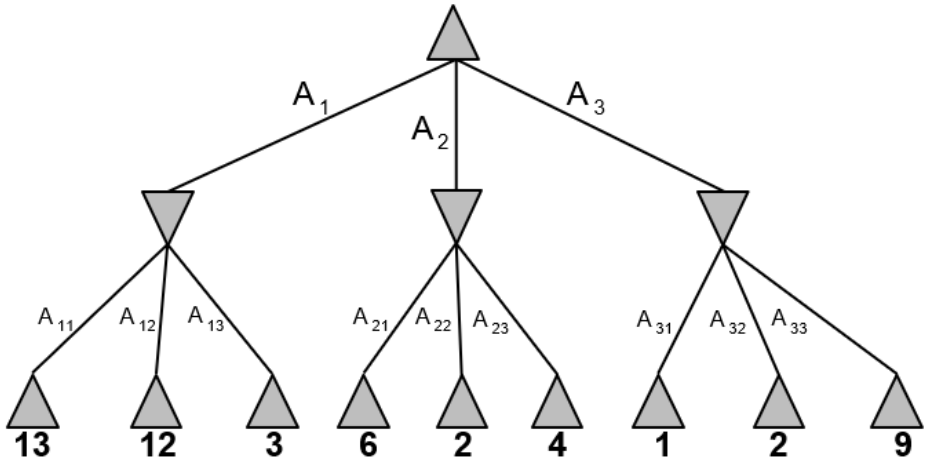
Tutorial Week 5-6

This weeks tutorial will cover both Adversarial Games and Knowledge representation and reasoning.

- Review definitions.
- Game trees and alpha-beta pruning.
- Explore the minimax and alpha-beta pruning algorithms.
- Backward/forward chaining
- Resolution example

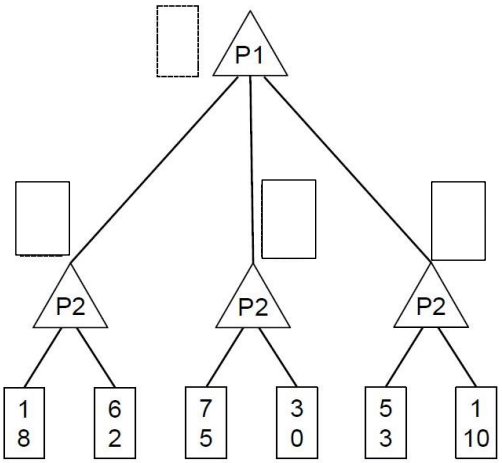
1) Define in your own words the following terms: zero sum game, game tree, pruning, terminal state, strategy, minimax value, utility function.

2) Fill in the values in each of the nodes in the following game tree. Then answer which branches in this game tree will be pruned by the Alpha-Beta algorithm.



3) Examine the following game tree description and answer the questions below.

For the following game tree, each player maximizes their respective utility. Let x, y respectively denote the top and bottom values in a node. Player 1 uses the utility function $U_1(x, y) = x$.



Both players know that Player 2 uses the utility function $U_2(x, y) = x - y$.

a) Fill in the rectangles in the figure with the pair of values returned by each max node.

b) Assume the branches are explored from left to right. Will any branches be pruned?

Knowledge representation and reasoning

4) Define in your own words the following terms: Logic, inference, model, satisfiability, validity, completeness, entailment, propositional logic, first-order logic, resolution, existential/universal quantification, forward/backward chaining, declarative vs procedural.

5) Convert the following set of sentences to clausal form.

S1: $E \Rightarrow D$.
S2: $C \wedge F \Rightarrow \neg B$.
S3: $A \Leftrightarrow (B \vee E)$.

6) Resolution.

Consider the following sentence:

$[(\text{Study} \Rightarrow \text{Pass}) \vee (\text{Persistence} \Rightarrow \text{Pass})] \Rightarrow [(\text{Study} \wedge \text{Persistence}) \Rightarrow \text{Pass}]$

a) Use truth tables to determine whether this sentence is valid, satisfiable, or unsatisfiable.
b) Convert the left-hand and right-hand sides of the main implication into CNF, showing each step, and explain how the results confirm your answer to (a)
c) Prove your answer to (a) using resolution.
d) Is there a contradiction in the clauses of question 5?

7) **Download the Tut5search.ipynb file. Start jupyter lab and work through the notebook.**

End